

CSA0611 ARTIFICIAL INTELLIGENCE

LAB PROGRAM 1

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AIM

To implement the **8-Queen Problem** using the **Backtracking algorithm** in Python and place eight queens on an 8×8 chessboard so that no two queens attack each other.

PROCEDURE

1. Start with an empty 8×8 chessboard.
2. Place a queen in the first column at a safe position.
3. Check that no queen exists in the same row, upper-left diagonal, or lower-left diagonal.
4. If the position is safe, place the queen and move to the next column.
5. If no safe position is available, backtrack to the previous column and try another position.
6. Repeat the process until all eight queens are placed successfully.
7. Display the final arrangement of the queens.

Algorithm – 8-Queen Problem

1. Start.
2. Create an empty 8 × 8 chessboard.
3. Begin placing queens from the first column.
4. Check if the current position is safe:
 - No queen in the same row.
 - No queen in the upper-left diagonal.
 - No queen in the lower-left diagonal.
5. If the position is safe, place the queen.
6. Move to the next column and repeat the process.
7. If no safe position is found, backtrack to the previous column and try another position.
8. Continue until all 8 queens are placed.
9. Display the final chessboard.
10. Stop.

Pseudocode – 8-Queen Problem

BEGIN

Create an empty 8 × 8 board

```

FUNCTION Solve(Column)

    IF Column = 8 THEN
        RETURN TRUE

    FOR each Row from 0 to 7

        IF Position(Row, Column) is Safe THEN

            Place Queen

            IF Solve(Column + 1) = TRUE THEN
                RETURN TRUE
            ENDIF

            Remove Queen
        ENDIF

    ENDFOR

    RETURN FALSE

END FUNCTION

Call Solve(0)

Display the board

```

Observation

The Backtracking algorithm successfully places eight queens on the chessboard such that no two queens attack each other. Whenever a conflict occurs, the algorithm backtracks and tries another position until a valid solution is found.

Result

The **8-Queen Problem** was successfully implemented in Python using the **Backtracking algorithm**, and a valid arrangement of eight queens was obtained.

```

1 N = 8
2
3 board = [[0 for i in range(N)] for j in range(N)]
4
5 def isSafe(row, col):
6     for i in range(col):
7         if board[row][i] == 1:
8             return False
9
10    i = row
11    j = col
12    while i >= 0 and j >= 0:
13        if board[i][j] == 1:
14            return False
15        i -= 1
16        j -= 1
17
18    i = row
19    j = col
20    while i < N and j >= 0:
21        if board[i][j] == 1:
22            return False
23        i += 1
24        j -= 1
25
26    return True
27
28 def solve(col):
29     if col >= N:
30         return True
31
32     for i in range(N):
33         if isSafe(i, col):
34             board[i][col] = 1
35
36             if solve(col + 1):
37                 return True
38
39             board[i][col] = 0
40
41     return False
42
43 if solve(0):
44     print("Solution:")

```

Enter Input here

If your code takes input, add it in the above box before running.

Output

Status : Successfully executed

Time:	Memory:
0.0100 secs	8.8 Mb

Your Output

Solution:

```

[1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0]
[0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1]
[0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0]

```